

Using Flutter for Cross-Platform Mobile Application Development

This article explains the features of the Flutter cross-platform mobile app development framework and how to configure it before getting started with development. It is aimed at students, college faculty members, researchers and developers interested in cross-platform mobile application development.



Building creative and user friendly mobile applications is a challenging and difficult task. There are various popular frameworks available that provide a variety of promising features for developing mobile applications. For instance, Android provides a native framework based on Java and the Kotlin language for mobile app development, while in the iOS framework, Objective-C or the Swift language is used. Hence, to develop applications for Android and iOS, we require different

frameworks and languages. Thankfully, there are now a number of frameworks that support mobile operating systems, with which developers can overcome this complexity when creating Web applications as well as desktop apps.

Frameworks that provide such multi-platform development support are known as cross-platform development tools. Using cross-platform development frameworks, it is possible to write code once and deploy it across various platforms like Android, iOS and on desktops. These are efficient in

terms of time and development efforts. The various tools include Ionic from Drifty, Xamarin from Microsoft, React Native from Facebook and Phonegap from Adobe. Recently, Google has introduced its new cross-platform application development tool named Flutter. Figure 1 highlights the features of the Flutter framework.

Flutter is a mobile application development SDK which can create apps that target both Android and iOS platforms simultaneously, using a single code base. Unlike other frameworks,